Understanding the Research Problem

Unit I · Lecture I.1 — Meaning, significance, and

why a *task*, a *project* and a *research problem* are three different animals

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Where we are, and what changes today

Lecture 0	I.1 today	I.2–I.3	I.4–I.6	I.7–I.8
why research	what *is* a problem	find & narrow one	judge & scope it	approach & data

What you already know how to do

Build things. You have shipped mini-projects, trained models, deployed apps. **You are good at execution.**

What today adds

Deciding *what deserves to be executed* — and being able to defend that decision to a reviewer.

Learning objectives for this hour

LO1 · Define
State what a research problem is, in your own words, without quoting a textbook.

LO2 · Distinguish
Given any piece of work, classify it as a task, a project or a research problem — and justify it.

LO3 · Diagnose
Spot the four commonest defects in a badly written problem statement.

LO4 · Convert
Take a build-task you already have and re-cast it as a researchable problem.

Maps to CO1 *Identify and formulate research problems using defined criteria and characteristics.*

Assumed background: data structures, algorithms, DBMS, a working knowledge of AI/ML, and at least one mini-project you have actually built.

Hook: a 90%-accurate model that failed

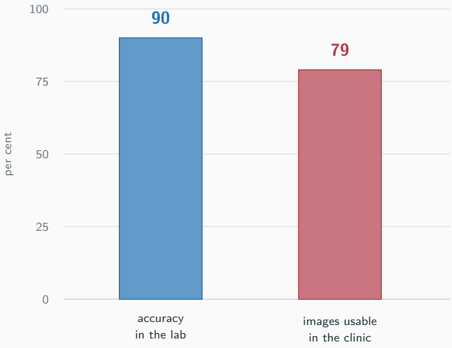
2020 · Google Health takes a diabetic-retinopathy AI to Thailand

The system. A deep model that reads retinal photographs and flags diabetic retinopathy. Published in *JAMA*. Lab accuracy on par with trained ophthalmologists — roughly **90%**.

The deployment. 11 clinics in Thailand. Nurses take the photograph, the model screens, patients avoid a specialist visit.

What happened. The system **rejected more than one in five images** outright. Clinic lighting was poor, so photographs were "low quality" by the model's standard. Internet was slow, so uploads timed out. Patients who had travelled for hours were told to come back.

Some nurses stopped using it.



Poll 1 · Hands up for exactly one option

Why did a 90%-accurate model fail in the field?

Choose the **single** best explanation. You must commit.

A

The model was overfitted. More training data would have fixed it.

B

The engineering was bad — slow internet and bad cameras are an implementation issue.

C

The *problem* was defined as "classify a clean retinal image" instead of "screen patients in a real clinic".

D

The nurses were not trained properly. It is a human-factors problem, not a research one.

The same project, two problem statements

What was actually solved

"Can a CNN classify diabetic retinopathy from retinal photographs **as accurately as an ophthalmologist**?"

Answer: yes. And it changed nothing for the patient in the queue.

What should have been asked

"Can an automated screening system **increase the number of patients correctly triaged per day** in a rural clinic with intermittent connectivity and uncontrolled lighting?"

Harder. Riskier. Actually useful.

The move that changed everything

The second statement carries its **constraints** inside it. "... *with* intermittent connectivity and uncontrolled lighting."

Discover: task, project, or research problem?

Think–Pair–Share 1 · Sort these six

1. Add dark mode to your college fest website.

2. Build a full attendance-management system with face recognition.

3. Fine-tune BERT on an Indian legal corpus and report F1.

4. Determine whether attention pruning degrades accuracy *more* on code-mixed Hindi–English text than on English.

5. Migrate the department database from MySQL to PostgreSQL.

6. Measure whether GitHub Copilot changes the *defect density* of code written by first-year students.

Think 90 s · Pair 2 min · Share 3 min

Label each one **T** (task), **P** (project) or **R** (research problem). Be ready to defend *one* of your labels.

The key that sorts all six

#	Item	Label	Why
1	Dark mode on the fest website	**T**	Method known, outcome known, hours of work
2	Face-recognition attendance system	**P**	Many tasks, months of work — but nothing *unknown*
3	Fine-tune BERT on legal text, report F1	**P**	Recipe exists. Becomes **R** the moment you ask *versus what, and why*
4	Pruning hurts code-mixed text more?	**R**	Outcome genuinely unknown; claim can be refuted
5	MySQL $\rightarrow$ PostgreSQL migration	**T**	Documented procedure, no new knowledge
6	Does Copilot change defect density?	**R**	Measurable, falsifiable, nobody knows the answer

Notice what did *not* decide the label

Not difficulty. Not how long it takes. Not whether it uses AI. **Only one thing: is the outcome already known?**

Now the definition — and it should feel like a summary

A research problem

A **research problem** is a clearly stated **gap between what is known and what needs to be known**, expressed so precisely that (i) an investigation can be designed to close it, and (ii) it is possible to be **wrong**.

SITUATION something in the world is unsatisfactory → **GAP** the literature does not tell us why or how → **QUESTION** sharp, bounded, answerable → **CLAIM** a statement that evidence could refute

Test If no possible experimental result would make you say "I was wrong", you do not have a research problem.

Task vs Project vs Research Problem

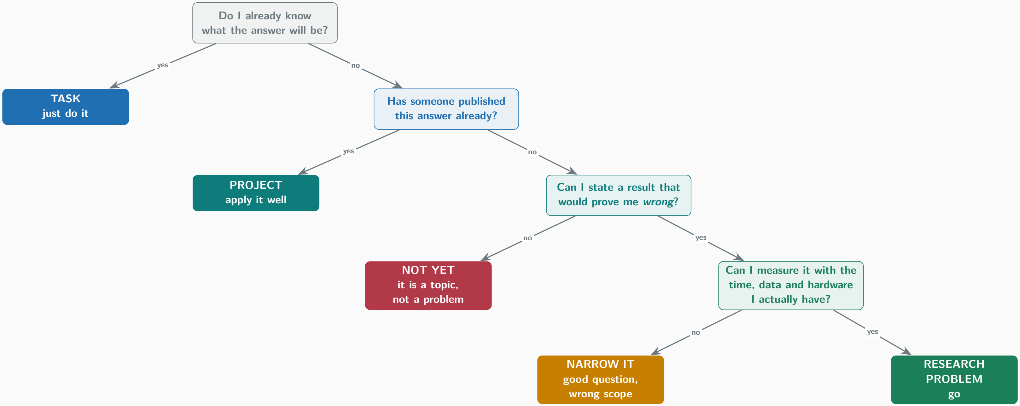
	TASK	PROJECT	RESEARCH PROBLEM
Starts from	An instruction	A requirement	A gap in knowledge
Unknown	Nothing	*How* to build it	*Whether* it holds at all
Output	A completed action	A working system	A defensible claim + evidence
Success	It is done	It works, on time	We now know something we did not
Failure	You did not do it	It does not work	**Still a result**, if explained
Ends when	The box is ticked	It ships	Someone else can reproduce it
Typical scale	Hours	Weeks to months	Months to years
Reviewer asks	"Is it done?"	"Does it work?"	"How do you know? Compared to what?"

DO IT

BUILD IT

KNOW IT

A decision tree you can run on your own idea



Homework tonight Run your own final-year idea down this tree. Note which node stops you.

The same distinction, inside real organisations

Organisation	A task there	A project there	A research problem there
Netflix	Add an artwork size	Rebuild the recommender service	Does personalised artwork change what a user actually *finishes*?
DeepMind	Add a logging hook	Port AlphaFold to new hardware	Can protein structure be predicted from sequence alone, to lab accuracy?
GitHub	Fix a Copilot UI bug	Ship Copilot for a new IDE	Does AI pair-programming change defect density and developer trust?
NPCI (UPI)	Add a bank to the switch	Build a merchant SDK	Can settlement stay correct and sub-second at 10^{10} txns/month?
ISRO	Recalibrate a sensor	Integrate the lander payload	Can a lander survive its own most likely failure modes?
Hospital IT	Export a DICOM batch	Deploy a PACS viewer	Does an AI second-read cut missed findings without more false alarms?

What the third column has in common

Every entry names a **measurable outcome** and at least one **constraint**. None of them names a technology.

Checkpoint quiz

Quiz · Write your four answers before any reveal

Q1

"Application of blockchain in healthcare." This is:

(A) a research problem (B) a topic
(C) a hypothesis (D) a methodology

Q2

Which single addition most reliably turns a *project* into a *research problem*?

(A) a bigger dataset (B) a deeper model
(C) a baseline to compare against (D) a nicer UI

Q3

Your experiment shows your proposed method is *worse* than the baseline. This is:

(A) a failed project (B) a result worth reporting
(C) proof the idea was bad (D) a reason to change the metric

Q4

"Our system is better." The first thing a reviewer will ask is:

(A) which language did you use? (B) better than what, measured how?
(C) is the code open? (D) how long did it take?

Quiz · answers

Q	Ans	Why
1	**B**	A topic names an *area*. A problem names a *gap* plus a measurable outcome. "Blockchain in healthcare" cannot be wrong, therefore it cannot be researched.
2	**C**	A baseline is what makes a claim comparative, and therefore refutable. Bigger and deeper change the engineering, not the epistemics.
3	**B**	A negative result you can *explain* is knowledge. Changing the metric until you win is the definition of a research ethics violation (Unit III).
4	**B**	"Compared to what, measured how?" — the single most common question in peer review and in your viva. Prepare the answer before you start.

Score yourself 4/4 excellent · 3/4 solid · $\leq$2 re-read this deck before Lecture I.2.

Significance: why the statement decides everything

Everything downstream is determined by this one paragraph

PROBLEM statement → **Literature** which 100 papers you must read → **Method** proof, experiment or study → **Data** what you must collect → **Metrics** what counts as success → **Paper** what you can honestly claim

change the problem in month 5 and every box downstream is rebuilt

Cheap to change now

Rewriting the problem statement in **week 1** costs an afternoon.

Expensive to change later

Rewriting it in **month 5** costs the semester — and usually the paper.

Activity · Spot the mistake (4 statements, 4 defects)

S1

"To study the applications of artificial intelligence in agriculture."

S2

"To build a website that predicts stock prices accurately using LSTM."

S3

"To develop a novel deep-learning model that achieves the highest accuracy for brain-tumour detection."

S4

"To completely solve the problem of fake news on social media using NLP."

Pairs · 3 minutes

Each statement has **one dominant defect**. Name it in three words or fewer.

The four defects — and the repaired statements

	Defect	Repaired
S1	Topic, not a problem	"Does a smartphone-only leaf-disease classifier retain its accuracy on images captured by farmers in the field, versus curated lab images?"
S2	Artifact instead of question	"Does an LSTM trained on price history alone outperform a naive random-walk baseline on NIFTY-50 next-day direction, over 2019–2024?"
S3	Unfalsifiable superlative	"Does adding an attention gate improve Dice score over U-Net and nnU-Net on BraTS-2021, at equal parameter budget?"
S4	Impossible scope	"Can source-credibility features improve fake-news detection F1 over text-only baselines, on Hindi-language WhatsApp forwards?"

The repair pattern is always the same

Add a **baseline** ("versus ..."), a **dataset**, a **metric**, and a **constraint**. Then it can be wrong — and therefore it can be research.

Guided practice

Worked example · a task becomes a research problem

Start (task).
"Build a chatbot for our college website."

+ **Baseline.** versus the existing FAQ search page.

+ **Metric.** % of queries resolved without a human, and time to resolution.

+ **Constraint.** on a 4 GB GPU, answering in Hindi and English.

= **Research problem.**
"Does a retrieval-augmented 7B model resolve more student queries without human escalation than keyword FAQ search, for code-mixed Hindi–English queries, under a 4 GB memory budget?"

```python
# The problem statement DICTATES the experiment.
# Read it once more, then read this code.

BASELINE = "keyword_faq_search" # <- from "versus"
SYSTEM = "rag_7b_quantised" # <- from the question
DATASET = "1200 real student queries, code-mixed" # <- constraint
METRICS = ["escalation_rate", # <- from "resolve without human"
           "time_to_resolution_s"]
BUDGET_GB = 4 # <- hard constraint

def run(system, dataset, metrics, budget_gb):
    assert peak_memory(system) <= budget_gb # constraint is TESTED
    return {m: evaluate(system, dataset, m) for m in metrics}

results = {name: run(name, DATASET, METRICS, BUDGET_GB)
          for name in (BASELINE, SYSTEM)}

# H0: escalation_rate(SYSTEM) >= escalation_rate(BASELINE)
# A result that fails to reject H0 is still a finding.
significant = paired_bootstrap(results, metric="escalation_rate", alpha=0.05)
```

Every named variable in the code traces back to one clause of the problem statement. That is the test of a good statement.

Your turn · convert one of *your* three problems

1. Your task
"I want to build . . ."
→
2. Versus what?
name a baseline
→
3. Measured how?
name one metric
→
4. Under what limit?
name one constraint
→
5. Rewrite
one sentence

Individual · 3 minutes · then 2 volunteers read theirs aloud

Use the sentence frame:

"Does ______________ improve ______________ over ______________, under ______________?"

Keep this sentence. Lecture I.4 will test it against formal criteria; Assignment 1 will be built on it.

Close

Reflection and key takeaways

Answer these in your head

1. What did we learn?
2. What surprised you most?
3. Where else could you apply the *"compared to what?"* question — outside research?

1 · A problem is a gap, not a topic

If it cannot be wrong, it cannot be researched.

2 · Task / Project / Research differ by one thing

Not difficulty, not duration, not technology — only whether the outcome is already known.

3 · Four ingredients repair almost anything

Baseline · dataset · metric · constraint.

4 · The statement is the cheapest thing to fix

An afternoon in week 1, or a semester in month 5.

Homework, reading and what comes next

Homework — due before Lecture I.2

H1. Run your final-year idea down today's decision tree. Write which node stopped you, in one line.

H2. Submit one problem statement using the sentence frame (*Does X improve Y over Z, under C?*).

H3. Find *one* published paper whose title contains a comparison. Copy its problem statement from the abstract.

Discussion questions to bring back

Is a replication study research? Is a benchmark paper research? Can a Kaggle winner be research?

Further reading

Textbook. Melville & Goddard, *Research Methodology for science & engineering students*, Ch. 1–2. Ganesan, *Research Methodology for Engineers* — problem identification.

Papers, 30 minutes each.
Beede et al., *A Human-Centered Evaluation of a Deep Learning System Deployed in Clinics*, CHI 2020 — today's hook.

Ginsberg et al., *Detecting influenza epidemics using search engine query data*, Nature 2009 — then the 2014 critique.

Shaw, *Writing Good Software Engineering Research Papers*, ICSE 2003.

Next · Lecture I.2 *Where research problems actually come from* — literature gaps, practice, replication and industry pain.

A problem well stated
is a problem half solved.

— Charles Kettering